

The empirical recurrence for OEIS A234986: recovering a conjecture that is not written down

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Abstract

OEIS A234986 records “Empirical recurrence of order 91”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 750$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 91, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 91 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234986 is “Number of $(n+1) \times (3+1)$ 0..5 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 2, with no adjacent elements equal (constant-stress tilted 1×1 tilings)”. It has offset 1 and begins

1640, 6908, 34608, 206556, 1303920, 8616068, 58209538, 400984904, ...

The entry states:

Empirical recurrence of order 91 (see link above).

As of the “Last modified” line on the live entry (Jun 20 2022 01:34:52, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 750$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 750$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1500$ exact terms of the model returns order 91 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{91} c_i a(n-i),$$

c_1	17	c_{24}	9528728482479180	c_{47}	−2971649659601894693093	c_{70}	33969184110
c_2	58	c_{25}	67873145185163549	c_{48}	−1428868631053309942886	c_{71}	−2501640500
c_3	−2293	c_{26}	−44434021371802625	c_{49}	3462499277925778099105	c_{72}	−760679594
c_4	1691	c_{27}	−368259051052501187	c_{50}	1740063499440481587495	c_{73}	5324668305
c_5	149075	c_{28}	165349645751992890	c_{51}	−3537437472180639466285	c_{74}	1417010752
c_6	−305378	c_{29}	1715480027663290052	c_{52}	−1823615452091257336246	c_{75}	−948630280
c_7	−6239638	c_{30}	−465762847303446470	c_{53}	3169030913521147659135	c_{76}	−216828178
c_8	16014478	c_{31}	−6885877706693832796	c_{54}	1648935038702526756395	c_{77}	1395649290
c_9	189042854	c_{32}	821278093925832995	c_{55}	−2488647266543885000011	c_{78}	268145938
c_{10}	−510981046	c_{33}	23892154894877115580	c_{56}	−1287950717054100656891	c_{79}	−16666423
c_{11}	−4408100060	c_{34}	254312641183542491	c_{57}	1711840700306205311201	c_{80}	−26233255
c_{12}	11520714443	c_{35}	−71860242619082220646	c_{58}	869112121237274953674	c_{81}	15794255
c_{13}	81973766342	c_{36}	−8691735039306670244	c_{59}	−1030061602884692859426	c_{82}	19720323
c_{14}	−195822964076	c_{37}	187812101086806584349	c_{60}	−506261895998428088540	c_{83}	−1152043
c_{15}	−1242537422252	c_{38}	40199511847038605982	c_{61}	541172965996266893926	c_{84}	−1092476
c_{16}	2600781395503	c_{39}	−427451042028586596565	c_{62}	254122087623714510902	c_{85}	6191184
c_{17}	15571354269015	c_{40}	−124786490049076726746	c_{63}	−247599051390091197226	c_{86}	4178552
c_{18}	−27589957649901	c_{41}	848742429299861586027	c_{64}	−109626062765293189324	c_{87}	−229057
c_{19}	−162951081657115	c_{42}	302411993723652797646	c_{65}	98313167466216268624	c_{88}	−98095
c_{20}	237024619886230	c_{43}	−1472550978998482034376	c_{66}	40493252829118728764	c_{89}	51668
c_{21}	1434832469521208	c_{44}	−601594741297753185412	c_{67}	−33729906654462029448	c_{90}	10616
c_{22}	−1661876795603836	c_{45}	2235271473988505344693	c_{68}	−12744996318337086336	c_{91}	−5308
c_{23}	−10696688346731271	c_{46}	1005305699178602784141	c_{69}	9943832229767324432		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 91, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $91 + 4$ terms removed returns order 91 as well, so $\deg q_{\min} = 91 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $91 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 91 that this sequence satisfies — and that family turns out to have exactly one member.

References

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